

Forecasting ICE Coffee C Futures: A Ten-Model Benchmark, Tests of the Random-Walk Hypothesis, and a Deployed GJR-GARCH Volatility Model

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Abstract—We benchmark ten forecasting models ranging from simple baselines to an AI foundation model on 32 years of ICE *Coffee C* nearby-close futures prices (1994–2026, 8,104 trading days). A rolling-window backtest over 60 evenly-spaced forecast origins, each with a 1,536-day context and a 63-day horizon, finds that a random walk with drift—one of the simplest models in the suite—attains the lowest mean absolute error (MAE). The Hansen-Lunde-Nason Model Confidence Set retains nine of ten models at $\alpha = 0.10$, eliminating only Prophet; pairwise Diebold-Mariano tests reach the same joint conclusion once corrected for multiple comparisons. To explain this tie, we run an *a priori* forecastability battery on the same series: stationarity tests (ADF, KPSS, Phillips-Perron), return autocorrelation (Ljung-Box), the heteroskedasticity-robust Lo-MacKinlay variance ratio, the Hurst exponent under Lo’s modified R/S, spectral predictability, permutation entropy, and the Rosenstein Lyapunov exponent with a shuffled-return surrogate null. All diagnostics agree the series is consistent with a near-random walk. Because mean-only models are statistically indistinguishable, we select the deployment model on the basis of *prediction intervals*: a GJR-GARCH(1,1)-*t* specification with constant mean and Student-*t* innovations, justified by AIC/BIC, residual diagnostics, and an expanding-window coverage check at 60 historical origins, in which empirical 80% and 95% intervals meet or slightly exceed nominal across the full horizon.

Index Terms—commodity forecasting, GARCH, foundation models, Model Confidence Set, market efficiency, prediction intervals

I. INTRODUCTION

Coffee is one of the world’s largest agricultural commodities and the ICE *Coffee C* contract is the international benchmark for arabica futures. Specialty roasters use the contract as the reference price for green-coffee purchases, so where the contract is headed over the months ahead is a question they have a practical interest in. The work reported here was carried out for 19 Coffee Company, a Pittsburgh specialty roaster, and aims to give an honest assessment of how predictable arabica futures actually are on a multi-month horizon, and—given that assessment—how to construct a forecast that is useful for procurement.

The work proceeds in three parts.

The first is a benchmark. We assemble ten forecasting models that span the methodological landscape, from a naive last-value forecaster to an AI foundation model (IBM Granite TTM, applied zero-shot) [1], and run them through a rolling-window backtest at 60 evenly-spaced forecast origins over

32 years of daily prices. The Hansen-Lunde-Nason Model Confidence Set [4] retains nine of the ten models at $\alpha = 0.10$; only Prophet [5] is eliminated. Pairwise Diebold-Mariano tests against the random-walk baseline [2], [3] reject Prophet strongly ($p < 10^{-3}$) and additionally flag Granite, Ridge, and (marginally) GARCH at conventional levels; once the nine pairwise comparisons are corrected via Bonferroni only the Prophet rejection survives, so DM and MCS reach the same joint conclusion. A pre-trained foundation model with a six-year context window does not beat a constant-drift extrapolator on point accuracy across the 60-origin evaluation.

*The second part is an *a priori* forecastability post-mortem.* The tie raises a natural question: is this an artifact of the model suite—a different class of models might extract structure ours misses—or does the series itself carry little exploitable structure? We answer with a battery of diagnostics applied to the series alone, independent of any forecasting model. We give primary weight to the standard finance-econometrics battery laid out in the Tsay [6] framework—ADF [7], KPSS [8], and Phillips-Perron [9] unit-root tests; Ljung-Box [10]; the Lo-MacKinlay variance ratio [11]; and the Hurst exponent [12] under Lo’s modified R/S correction [13]—and supplement it with three exploratory metrics from the nonlinear-dynamics and signal-processing literatures (spectral predictability [14], permutation entropy [15], the Rosenstein Lyapunov exponent [16] with a shuffled-return surrogate null [17]). Both halves converge on the same answer. The empirical fingerprint is consistent with the weak form of market efficiency [18]: past prices alone do not contain information about future prices that any of these models can extract.

The third part is the deployment model. If mean-only point forecasts are indistinguishable, the deployment model should be chosen on the basis of *prediction intervals* rather than mean MAE. Coffee returns exhibit pronounced volatility clustering and heavy tails, both of which a constant-variance interval ignores. We fit a GJR-GARCH(1,1) model with constant mean and Student-*t* innovations [19], [20], justify each specification choice with a statistical test, and validate the resulting intervals through an expanding-window backtest in which empirical 80% and 95% coverage meet or slightly exceed nominal across the full 63-day horizon.

The remainder of the paper is organized as follows. Section II describes the project’s client context. Section III

introduces the dataset. Section IV defines the forecasting suite and the backtest design. Section V reports the rolling-window backtest results. Section VI reports the DM and MCS statistical tests. Section VII reports the a priori forecastability diagnostics. Section VIII specifies and validates the GJR-GARCH(1,1)-t deployment model. Section IX sketches the live deployment pipeline. Section X concludes.

II. PROJECT CONTEXT: 19 COFFEE COMPANY

19 Coffee Company is a small-batch specialty coffee roaster based in the South Hills of Pittsburgh. The business imports unroasted *green coffee* from origin, roasts it in small batches, and delivers the roasted product to local cafes, restaurants, and gourmet grocers across the tri-state area. Its sourcing philosophy emphasizes directly-traded and relationship-based purchases, fair-trade and organic certifications, and shade-grown beans cultivated without pesticides or herbicides, with long-standing partnerships including Building New Hope in Central America and the worker-owned Firsthand Cooperative in Appalachia.

For a specialty roaster of this scale, the ICE *Coffee C* contract is the industry-standard reference price for arabica green coffee. Even when a roaster does not transact directly on the futures exchange, the Coffee C settlement is the benchmark against which directly-traded and relationship-based green-coffee purchases are priced, with origin-specific differentials added on top. The forward distribution of Coffee C is therefore the right object for a procurement-side forecast: not a single point estimate of where prices will be in three months, but a calibrated forward range that quantifies how wide the plausible price envelope is around that point. The benchmark in Sections V–VI establishes which classes of models can be expected to contribute beyond a naive baseline; the diagnostics in Section VII explain why that set is narrower than one might expect on a 32-year series; and the GJR-GARCH deployment in Section VIII produces the calibrated range itself.

III. DATA

The series is the ICE *Coffee C* nearby-close (front-month, most actively traded) daily settlement from 3 January 1994 through 30 January 2026, totaling 8,104 trading days. The raw source is the ICE historical-prices export [21]; we retain only the date column and the front-month close, drop missing rows, and persist the result as a two-column CSV. All subsequent analysis works from this cleaned series.

Prices are quoted in US cents per pound. The Coffee C contract trades in lots of 37,500 lb (≈ 250 60-kg bags), so every error metric in this paper inherits the cents-per-pound scale—an MAE of 12.8 cents/lb corresponds to roughly \$4,800 per contract on average.

Fig. 1 shows the price history. Three features are worth noting in advance. First, the series spans multiple distinct regimes, including the 1997 Brazilian frost-driven peak, the 2010–2011 rally, the 2014 drought-driven spike, and the 2024–2025 supply-disruption rally that pushed prices above 400 cents/lb for the first time in the record. Second, the

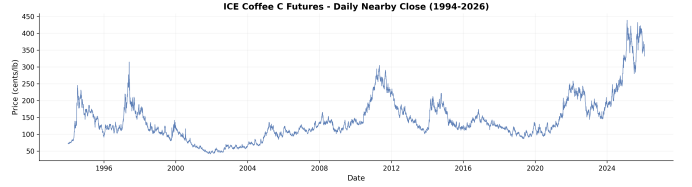


Fig. 1. ICE *Coffee C* nearby-close daily settlement, 1994–2026 (8,104 trading days). Major regime episodes are visible at the 1997 Brazilian frost, the 2010–2011 rally, the 2014 drought spike, and the 2024–2025 rally past 400 cents/lb.

long-run trajectory is non-stationary; we work with log-returns wherever stationarity is required. Third, the visible roughness of the price path—periods of rapid swing concentrated in some regimes (1994–1997, 2011, 2024) and quieter, smoothly-trending stretches in others (2003–2005, 2017–2019)—is the volatility-clustering pattern that motivates a GARCH-family model in Section VIII; the formal diagnostic for conditional heteroskedasticity belongs to the log-returns plot in Fig. 6 rather than to the price level shown here.

IV. FORECASTING SUITE AND BACKTEST DESIGN

A. Model suite

Table I lists the ten models. The suite is constructed to span the historical landscape of point-forecasting methods, from the simplest baselines through classical statistical models, a volatility-aware extension, a structural-decomposition method, recursive machine-learning approaches, and a contemporary AI foundation model. Within each category we selected a well-established representative rather than tuning across alternatives, so the benchmark tests whether any *family* offers a systematic advantage over the others rather than whether a particular instance can be hyperparameter-optimized to win. Grouping the ten models into five categories also makes the methodological breadth easier to read than a flat ten-row list. All models share a uniform interface: each is wrapped in a class exposing `predict(context, horizon)` returning a point forecast at every step of the horizon. Stochastic components (RandomForest at construction time and any incidental randomness in feature construction) are seeded; the rest of the suite is deterministic given fixed inputs (statsforecast and arch use maximum-likelihood estimation, Granite TTM is zero-shot, Prophet uses maximum a posteriori estimation). All implementations rely on standard Python scientific libraries: statsforecast [22], arch [23], Prophet [5], scikit-learn [24], statsmodels [25], and the IBM tsfm toolkit [1].

The *GARCH* entry in Table I is a GARCH(1,1) model with an AR(1) mean and Normal innovations fitted on price levels, used here as a point-forecast wrapper alongside the rest of the suite. The GJR-GARCH(1,1)-t model that Section VIII specifies for the deployment is a separately-specified variant fitted to log-returns with Student-*t* innovations. The two models share a family name but address different questions: the former contributes a point forecast to the benchmark, the latter produces calibrated prediction intervals for deployment.

TABLE I
FORECASTING SUITE

Category	Model
Simple baselines	Naive, Random Walk with Drift (RWD), Theta [26]
Classical statistical	ETS [27], ARIMA (auto-selected) [28], [29]
Volatility	GARCH(1,1) with AR(1) mean [19]
Structural	Prophet [5]
Recursive ML	Ridge [30], Random Forest [31]
Foundation (zero-shot)	IBM Granite-TimeSeries-TTM-R2 [1]

B. Backtest protocol

We use a rolling-window backtest with three design parameters:

- **Context window:** 1,536 trading days (≈ 6 years). The Granite-TimeSeries-TTM-R2 release ships several zero-shot context-length variants (512, 1024, 1536); we use the largest—specifically the `1536-96-r2` variant, with a native 96-step prediction length trimmed to the 63-day forecast horizon—so that every model in the suite is given as much history as possible while keeping the comparison fair.
- **Horizon:** 63 trading days (≈ 3 months), long enough to be operationally relevant for green-coffee procurement and short enough that point forecasts remain defensible.
- **Origins:** 60 evenly-spaced forecast origins distributed across the 32-year history. The choice of 60 emerged from a small sweep (1, 10, 30, 60); the cluster-vs-outlier structure of the leaderboard is stable from 30 to 60 origins, and within-cluster rankings shuffle by one to three positions while all cluster-mean MAE values remain within roughly 0.3 cents/lb of each other in both cases.

At each origin a , every model receives the same 1,536-day context $\{y_{a-1535}, \dots, y_a\}$ and produces a 63-step forecast. We compute five forecast-error metrics per (origin, model): MAE, RMSE, MAPE, sMAPE, and MASE, the last following the scale-independent definition of Hyndman and Koehler [32]. The resulting tables hold 1,010 rows (101 origins across the four scales $\times$ 10 models) at the summary level and 63,630 rows (origins $\times$ models $\times$ horizon steps) at the per-step level. The per-step table is the input to the Diebold-Mariano test and the Model Confidence Set in Section VI. No model sees test data during fitting; every origin gets a fresh fit on the preceding 1,536 days.

V. BACKTEST RESULTS

A. Qualitative preview: a single recent window

Before turning to the 60-origin evaluation, Fig. 2 shows the 10-model overlay on the most recent 63-day window of the series, with the realized price path in black. Every model produces a visually plausible trajectory, and several track the realized path quite closely on this particular window—including Granite TTM, which has the lowest MAE here.

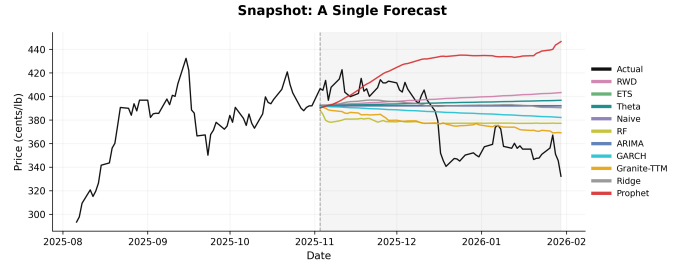


Fig. 2. Qualitative preview: 10-model forecasts on the most recent 63-day window, overlaid against the realized price path (black). The shaded region marks the held-out forecast horizon. Every model is visually plausible on this single window; the 60-origin evaluation that follows tells the rigorous story.

TABLE II
60-ORIGIN MEAN ERROR METRICS, SORTED BY MAE

Model	MAE	RMSE	MASE	sMAPE
RWD	12.79	14.84	6.10	0.083
ETS	12.82	14.89	6.11	0.085
Theta	12.86	14.93	6.13	0.085
Naive	12.91	14.99	6.15	0.085
RF	13.13	15.19	6.28	0.087
ARIMA	13.24	15.39	6.28	0.085
GARCH	13.51	15.74	6.50	0.090
Granite-TTM	13.84	16.09	6.55	0.092
Ridge	14.68	17.16	6.87	0.101
Prophet	22.69	24.99	10.60	0.152

The figure is intended as a sanity check, not as evidence; the formal across-origin evaluation in Sections V.B–D returns a very different ranking from what this single window suggests, and the disagreement is itself a useful warning about reading too much into any one window.

B. Leaderboard at 60 origins

Table II reports mean per-origin error metrics at the 60-origin scale, sorted by MAE. The arithmetic leader is the random walk with drift, followed by the other simple baselines (ETS, Theta, Naive) within 0.15 MAE of it. The two machine-learning models (Random Forest, Ridge) and the AI foundation model (Granite TTM) all rank below the simple baselines on mean MAE; increased model complexity does not translate into lower point-forecast error on this series. Prophet sits well outside the cluster.

C. Distribution of per-origin MAE

Mean metrics sort the models but can hide right tails; a model with average behavior most of the time but occasional collapses will look worse than it deserves, and vice versa. Fig. 3 shows the full per-origin MAE distribution at scale 60. Boxes are sorted by median, the mean is overlaid as a black diamond, and the nine indistinguishable cluster models are colored neutral with Prophet in red.

Two patterns are visible. First, the IQRs of the nine non-Prophet models overlap heavily—visual evidence for the statistical tie formalized in Section VI. Second, the mean-minus-median gap for the nine cluster models is similar in magnitude

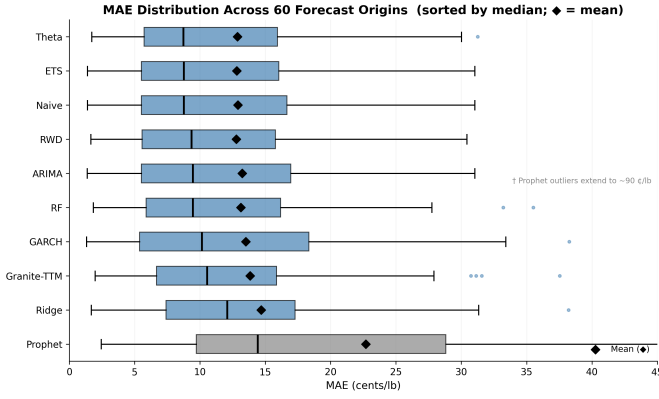


Fig. 3. MAE distribution across 60 forecast origins, sorted by median MAE (worst at top). Diamonds mark the mean. The nine MCS-retained models (steelblue) have heavily overlapping interquartile ranges; Prophet, the only MCS-eliminated model (gray), sits well to the right of the cluster with a much heavier right tail.

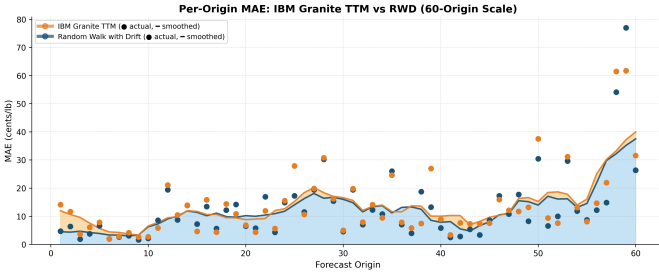


Fig. 4. Per-origin MAE: IBM Granite TTM versus RWD across 60 origins. Markers are observed values; lines are smoothed trends. Granite achieves lower MAE at only 16 of 60 origins; the orange band is the excess error above RWD.

(roughly 2.6–4.1 cents/lb), so no cluster model is distinguished by a particularly asymmetric distribution. Prophet’s mean-minus-median gap is 8.3 cents/lb, reflecting catastrophic tail behavior on a subset of origins rather than a uniformly higher error level.

D. Granite TTM versus RWD across origins

The boxplot in Fig. 3 already shows that Granite TTM has a higher *mean* and *median* MAE than RWD, with a mean-minus-median gap of 3.3 cents/lb that is in the middle of the cluster range. A natural follow-up is whether the underperformance is concentrated in a few collapsed windows or distributed across the historical record. Fig. 4 answers this directly by overlaying the per-origin MAE of Granite TTM and RWD at each of the 60 origins. The orange-shaded area marks origins where Granite exceeds RWD; the area is nearly continuous, and Granite achieves lower MAE at only 16 of 60 origins. The foundation model underperforms the drift baseline not because of a small number of collapses but through a steady positive bias across the historical record.

VI. STATISTICAL TESTS

The leaderboard rankings are descriptive and do not distinguish sampling noise from real differences in predictive

accuracy. We use two complementary tools to do so: the Diebold-Mariano test pairwise against the RWD baseline, and the Model Confidence Set as a joint test on the full suite.

A. Diebold-Mariano test versus RWD

The DM test [2] addresses the null that two forecasts have equal predictive accuracy. We use the per-origin MAE loss as the comparison statistic and estimate the HAC variance across origins with the Bartlett kernel and the Newey-West [33] rule-of-thumb bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$. We apply the Harvey-Leybourne-Newbold [3] small-sample correction: the statistic is multiplied by $\sqrt{(T+1-2h+h(h-1)/T)/T}$ (here $h = 1$, so by $\sqrt{(T-1)/T}$) and referenced against the t_{T-1} distribution rather than $\mathcal{N}(0, 1)$.

Three patterns appear. The top-tier simple models (ETS, Theta, Naive) do not differ significantly from RWD ($p > 0.4$); their gaps fall within sampling noise. Granite TTM is significantly worse than RWD at the 5% level (DM = +2.35, $p = 0.022$), and Ridge at the same level (DM = +2.04, $p = 0.046$). Prophet is rejected at the 0.1% level (DM = +4.53, $p < 0.001$). GARCH sits just over the 10% threshold (DM = +1.76, $p = 0.083$).

B. Model Confidence Set

DM compares each model to a fixed baseline and does not correct for multiple comparisons; running nine pairwise tests against RWD inflates the Type I error rate. The Model Confidence Set [4] addresses this by returning a *set* of models statistically indistinguishable from the best at significance α . The procedure starts with all ten models, computes the max- t statistic across centered loss differentials, bootstraps the null distribution (circular block bootstrap, $B = 5000$, block size 5), eliminates the model with the largest t -statistic if $p \leq \alpha$, and repeats until no further elimination is possible. p -values reported for eliminated models are monotonically adjusted across elimination steps, so an MCS p -value for a later-eliminated model cannot fall below that of an earlier-eliminated one.

At $\alpha = 0.10$, the MCS retains nine of the ten models: Naive, RWD, Theta, ETS, ARIMA, GARCH, Ridge, RF, and Granite-TTM. Only Prophet is eliminated, with MCS $p = 0.009$. The retained set spans every category in the suite, including the foundation model.

C. Reconciling DM and MCS

The two tests can look superficially contradictory: DM rejects equal accuracy between RWD and Granite (and Ridge) at the 5% level, while MCS keeps both in the confidence set with $p = 1.000$. The gap is real and reflects what each test does. DM tests one pair at a time without multiple-comparisons correction; with nine pairwise tests against RWD the family-wise error rate inflates. A Bonferroni adjustment to the pairwise level— $\alpha/9 \approx 0.006$ for an overall 5% rate—would eliminate the marginal Granite-TTM and Ridge significances entirely and leave only Prophet rejected, which is the conclusion the MCS reaches by construction. MCS performs the joint test

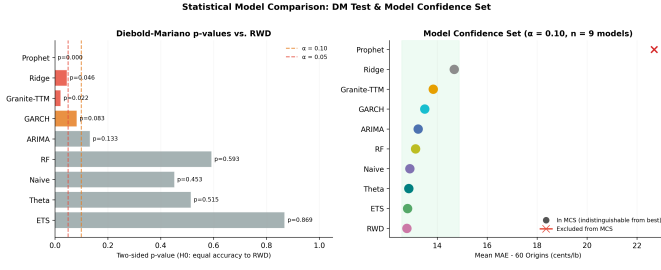


Fig. 5. Statistical model comparison. *Left*: Diebold-Mariano p-values versus RWD; bars colored by significance threshold ($p < 0.05$ in red, $p < 0.10$ in orange). *Right*: Model Confidence Set at $\alpha = 0.10$; survivors marked as filled circles, Prophet (the sole eliminated model) as a red cross. The green band shows the range of mean MAE across MCS survivors.

on the full set and corrects for multiple comparisons without requiring an explicit Bonferroni correction. With ten models the joint test is the appropriate primary tool; DM is a familiar pairwise summary that supplements rather than contradicts it.

A caveat on power. $T = 60$ origins is on the small side for MCS (the Hansen-Lunde-Nason simulations use $T = 100$ – 500). With $T = 60$ and ten models the test has limited power, so a wider-than-true MCS is partly a power story. The qualitative finding—Prophet eliminated, the remaining nine retained—is robust, but the surviving set could plausibly shrink with more origins.

D. Reading

Nine of ten models—spanning random-walk baselines, classical statistical models, a volatility model, recursive ML, and a foundation model—are jointly indistinguishable from the best at $\alpha = 0.10$. The leaderboard in Table II is therefore descriptive rather than a true ranking: RWD is not “the winner” but the arithmetic leader of a cluster of statistically-equivalent models, and the benchmark does not have enough resolving power to single one out. A diverse 10-model suite that includes a 32-year pretrained foundation model cannot statistically distinguish itself from a naive drift forecaster on this series.

VII. A PRIORI FORECASTABILITY DIAGNOSTICS

The benchmark tie has two possible readings. Either our suite is the wrong instrument and a different class of models would extract structure ours misses, or the series itself carries little exploitable structure at this horizon. We test the second reading directly with a battery of diagnostics applied to the series alone, independent of any forecasting model. The diagnostics split into a standard finance-econometrics battery (Sections VII-A–VII-D) and three exploratory metrics drawn from the nonlinear-dynamics and signal-processing literatures (Section VII-E).

A. Stationarity

Several of the diagnostics below (variance ratio, Hurst with R/S, the Lyapunov reconstruction) require a stationary input series. Fig. 6 shows the price level and its log-returns side by side. We confirm stationarity through three complementary

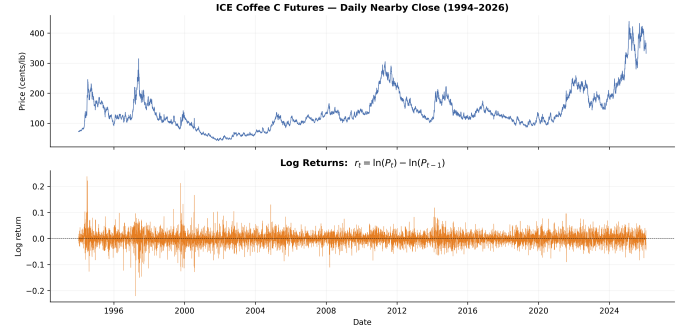


Fig. 6. (Top) ICE *Coffee C* nearby-close prices, 1994–2026. (Bottom) Daily log-returns $r_t = \ln P_t - \ln P_{t-1}$. The mean of r_t is $+1.9 \times 10^{-4}$ per day; visible heteroskedasticity around zero is the volatility-clustering pattern that motivates a GARCH-family model.

unit-root tests: the Augmented Dickey-Fuller [7] (H_0 : unit root); the KPSS [8] test (H_0 : stationary); and the Phillips-Perron [9] test (H_0 : unit root, non-parametric correction for serial correlation and heteroskedasticity).

All three tests agree. On prices, ADF fails to reject the unit-root null ($p = 0.557$), Phillips-Perron likewise fails to reject ($p = 0.418$), and KPSS rejects stationarity ($p < 0.01$). On log-returns the directions flip: ADF and Phillips-Perron strongly reject the unit-root null ($p < 10^{-90}$), and KPSS fails to reject stationarity ($p > 0.10$). The confirmatory design—rejection by ADF and Phillips-Perron paired with non-rejection by the oppositely-directed KPSS test—eliminates the standard “low-power ADF” critique. The Phillips-Perron agreement is particularly valuable here because its non-parametric correction is robust to the heteroskedasticity that Section VIII formally documents.

B. Return autocorrelation

Under the weak form of market efficiency, daily returns should not be linearly predictable from their own past. The Ljung-Box test [10] checks the joint null that all autocorrelations up to lag L are zero. With $T = 8,103$ log-returns and $L \in \{10, 20\}$, both windows formally reject ($p \approx 0.006$ at both lag choices). The magnitudes underneath this rejection are tiny: the mean of $|\hat{\rho}(k)|$ over lags 1–20 is approximately 0.013, with a maximum of 0.040. This is the textbook stylized fact for liquid financial returns [34]: linear return autocorrelations are statistically non-zero at very large T but economically negligible, too small to exploit through point-forecast models. The heteroskedasticity-robust variance-ratio test in Section VII-C provides a complementary direct test of the random-walk hypothesis with results that align with this practical reading.

C. Lo-MacKinlay variance ratio

Under the random-walk null, the variance of q -period returns equals q times the variance of one-period returns, so $VR(q) = \text{Var}(r_t^{(q)})/[q \cdot \text{Var}(r_t)]$ equals 1 at every horizon. Deviations are direct evidence against the random-walk hypothesis. We use the heteroskedasticity-robust z -statistic of Lo and MacKinlay [11], which handles the volatility clustering

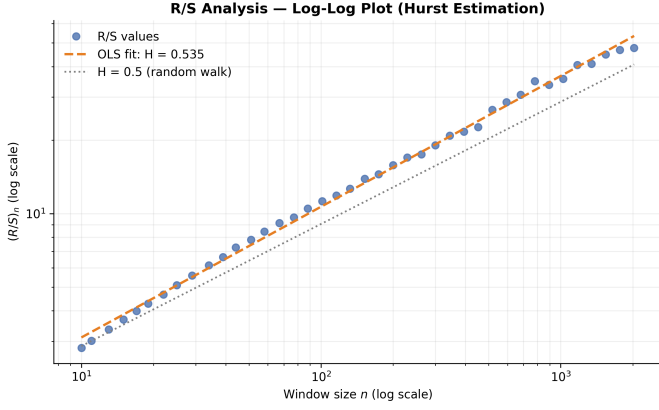


Fig. 7. Rescaled-range analysis of coffee log-returns. Empirical $(R/S)_n$ values (blue circles) and OLS log-log fit (orange dashed) yield $H = 0.535$. The gray dotted reference line is the random-walk slope $H = 0.5$. Lo’s modified $V_n = 1.137$ lies inside the 95% acceptance region $[0.809, 1.862]$, so the no-long-memory null is not rejected.

coffee returns exhibit. Across $q \in \{2, 4, 8, 16, 32\}$, $VR(q)$ stays within rounding distance of 1 (from 0.986 at $q = 2$ to 0.852 at $q = 32$) and no z -statistic reaches conventional significance. We fail to reject the random-walk null at every tested horizon. This is the strongest direct evidence in the a priori battery that the series is consistent with a random walk.

D. Hurst exponent with Lo’s modified R/S

The Hurst exponent H is estimated from the log-log slope of the rescaled range, $\mathbb{E}[(R/S)_n] \sim c \cdot n^H$ [12]. $H = 0.5$ corresponds to no exploitable memory; $H > 0.5$ to long-range persistence and $H < 0.5$ to mean reversion. Classical H is biased upward by short-range autocorrelation, so we also compute Lo’s modified R/S statistic V_n [13], which applies a Bartlett-kernel HAC adjustment. Under the long-memory null, $V_n \in [0.809, 1.862]$ at the 95% level. We compute both on log-returns, the standard representation for R/S analysis, and use the rule-of-thumb truncation lag $q = \lfloor T^{1/3} \rfloor$ ($q = 20$ at $T = 8,103$) as an alternative to the data-driven Andrews (1991) bandwidth [35]; V_n sits comfortably inside the acceptance region under both choices.

We find $H = 0.535$ and $V_n = 1.137$. The Hurst estimate sits just above the random-walk value, but Lo’s modified V_n lies inside the 95% acceptance region: we fail to reject the no-long-memory null. The Hurst-estimate excess of 0.035 above 0.5 is plausibly explained by the small short-range autocorrelations documented in Section VII-B, which classical H is known to absorb. Fig. 7 shows the underlying R/S log-log plot.

E. Exploratory metrics

To check the standard-battery conclusion from independent angles, we also computed three metrics from the nonlinear-dynamics and signal-processing literatures: spectral predictability Ω [14], Bandt-Pompe permutation entropy [15], and the Rosenstein largest-Lyapunov estimator [16] validated

with a shuffled-return surrogate-data null [17]. All three converge with the standard battery: spectral mass on log-returns sits at the white-noise floor ($\Omega \approx 0.05$, against 0.68 on price levels, where the apparent “structure” is the $I(1)$ drift signature rather than exploitable periodicity); permutation entropy sits near the maximum-randomness ceiling on both representations (0.97 on prices, ≈ 1.00 on log-returns); and the raw Lyapunov estimate ($\lambda = +0.071$) is indistinguishable from the shuffled-return surrogate distribution (one-sided $p = 0.97$), consistent with finite-sample bias of the Rosenstein algorithm on stochastic data rather than with deterministic chaos. Implementation details, the multi-frequency robustness check, and a discrete-length adaptation of Ω that replaces the original $\log(2\pi)$ denominator with $\log N_{\text{freq}}$ to keep the metric in $[0, 1]$ at long series, are available in the project repository [36].

F. Synthesis

The diagnostics agree. ADF and Phillips-Perron fail to reject the unit-root null on prices and strongly reject it on log-returns; KPSS, with the reversed null, agrees. Ljung-Box rejects formally on returns but the underlying autocorrelations are on the order of 10^{-2} , economically negligible. The heteroskedasticity-robust Lo-MacKinlay variance ratio fails to reject the random-walk null at horizons from 2 to 32 days. The Hurst estimate is 0.535 but Lo’s modified V_n does not reject the no-long-memory null. Spectral energy on log-returns is near the white-noise floor. Permutation entropy is near the ceiling on both representations. The raw Lyapunov exponent is consistent with what shuffled returns produce.

The empirical signature is consistent with weak-form market efficiency [18]: past prices alone do not contain information about future prices that simple baselines and complex models alike could extract on this series. The a priori prior is that any model in this class should be roughly competitive with a random-walk baseline; the benchmark in Sections V–VI confirmed that prior.

VIII. VOLATILITY MODEL FOR PREDICTION INTERVALS

If point forecasts from the suite are statistically indistinguishable, the deployment model must be chosen on a different basis. Coffee returns exhibit pronounced volatility clustering (Fig. 8) and heavy tails; both features make calibrated prediction intervals non-trivial. The constant-variance bands a simple model would produce are wrong in two directions at once—too wide in calm regimes, too narrow in stressed ones. We adopt a GJR-GARCH(1,1) model with constant mean and Student- t innovations as the deployment specification, following the standard framework for daily-frequency financial-return modeling [6]. Each modeling choice below is justified by a test on the same dataset; the section walks through each in turn.

A. Mean specification

A HAC-robust regression of log-returns on a constant returns a daily mean of $+1.88 \times 10^{-4}$ with Newey-West standard error 2.55×10^{-4} (10 lags), $t = +0.74$, $p = 0.46$. The drift is not statistically distinguishable from zero. We adopt a constant-mean specification.

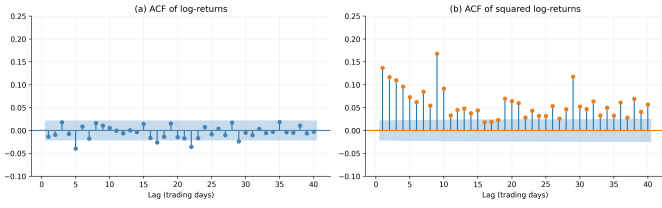


Fig. 8. ACF of log-returns (left) and squared log-returns (right) for coffee. Returns are nearly white noise; squared returns are persistently autocorrelated, the textbook fingerprint of conditional heteroskedasticity.

B. Innovation distribution

Daily log-returns have sample skewness $+0.19$ and excess kurtosis $+6.65$. Jarque-Bera [37] rejects normality at the limit of numerical precision ($JB = 1.5 \times 10^4$, $p < 10^{-300}$). A maximum-likelihood fit of the Student- t distribution to the raw returns yields $\nu \approx 4.2$, indicating substantially heavier tails than Normal. We use Student- t innovations.

C. ARCH versus IID

Engle’s ARCH LM test [38] with 10 lags yields $LM = 523.2$, $p \approx 5 \times 10^{-106}$. The null of no conditional heteroskedasticity is rejected at any reasonable significance level. Fig. 8 contrasts the ACF of returns with the ACF of squared returns: returns are nearly white noise, while squared returns are persistently autocorrelated out to dozens of lags—volatility clustering, the empirical pattern GARCH is designed to capture.

D. Symmetric versus asymmetric volatility

GARCH(1,1) [19] sets $h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}$. GJR-GARCH [20] adds an asymmetry term:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \mathbb{I}_{\{\varepsilon_{t-1} < 0\}} \varepsilon_{t-1}^2 + \beta h_{t-1},$$

allowing negative and positive shocks of equal magnitude to contribute differently to next-period variance.

Fitting both specifications to the full sample yields:

- **GARCH(1,1)-t**: AIC = 35492.7, BIC = 35527.7; Ljung-Box on squared standardized residuals $p_{LB(10)} = 0.0003$, $p_{LB(20)} = 0.0010$.
- **GJR-GARCH(1,1)-t**: AIC = 35450.4, BIC = 35492.4; Ljung-Box on squared standardized residuals $p_{LB(10)} = 0.048$, $p_{LB(20)} = 0.093$.

GJR improves AIC by 42 and BIC by 35; the Ljung-Box squared-residual p-values climb by roughly two orders of magnitude at both lag choices. $LB(10)$ remains just under 0.05, indicating some residual structure at short lags; $LB(20)$ clears 0.05. We adopt GJR-GARCH(1,1)-t as a defensible complexity/fit trade-off.

E. Fitted-parameter diagnostics

The fitted leverage parameter is $\gamma = -0.058$, statistically distinguishable from zero ($p < 0.001$). For equities, the GJR leverage parameter is typically positive: negative price shocks increase volatility more than positive shocks of equal size (the

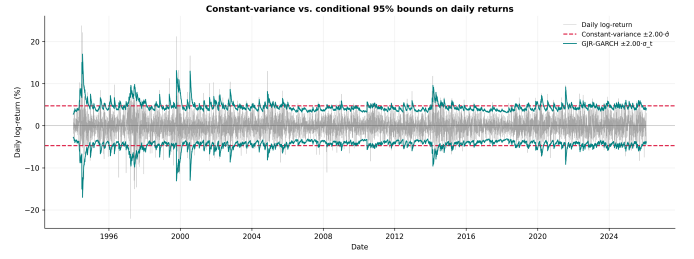


Fig. 9. Constant-variance $\pm 1.96\hat{\sigma}$ band (red, horizontal) versus GJR-GARCH conditional $\pm 1.96\sigma_t$ band (teal) on daily log-returns. Both achieve $\approx 95\%$ unconditional coverage; the conditional band adapts to regime, expanding during stress periods and narrowing during calm.

”leverage effect”). The sign on coffee is reversed: positive return shocks inflate volatility more than negative ones. This is plausible for soft commodities where supply-disruption events (frost, drought, geopolitical) coincide with the largest upward price jumps; ”good news for the price” and ”bad news for the supply” arrive together, and both inflate volatility.

Volatility persistence is $\alpha + \gamma/2 + \beta = 0.973$, corresponding to a half-life of volatility shocks of approximately 26 days—typical for daily commodity volatility, with persistence below 1 (the stability condition for unconditional variance to exist). The fitted Student- t degrees of freedom on standardized residuals is $\nu \approx 5.7$, higher than the $\nu \approx 4.2$ obtained from the raw-return fit in the distribution section. The two values are not in conflict: they measure different quantities. The raw-return ν captures the tail thickness of *unconditional* returns, which inherits heavy tails partly from volatility clustering itself; the residual ν captures the tail thickness of innovations *after* GJR-GARCH absorbs the time-varying variance. Once that variance is filtered out, the residual distribution looks closer to Normal, although still heavy enough that Student- t innovations are required.

F. Conditional versus constant-variance intervals

Fig. 9 contrasts the constant-variance $\pm 1.96\hat{\sigma}$ band against the GJR-GARCH conditional $\pm 1.96\sigma_t$ band, overlaid on the realized log-returns. Both bands achieve roughly 95% unconditional coverage in-sample, but the *when* differs. The constant band is too wide in calm regimes (2002–2005, 2017–2019), wasting precision when realized volatility is low, and too narrow in stressed regimes (1997 frost, 2010–2011 rally, 2024–2025 rally), failing to cover when realized volatility spikes. The conditional band expands and contracts with realized volatility, which is the difference between useful and misleading intervals for downstream procurement decisions.

G. Out-of-sample coverage

Specification tests show the model fits in-sample; the operational question is whether its prediction intervals are calibrated out-of-sample. We refit GJR-GARCH(1,1)-t at the same 60 evenly-spaced origins used in the Section V benchmark, this time on *expanding* windows (all history up to each origin).

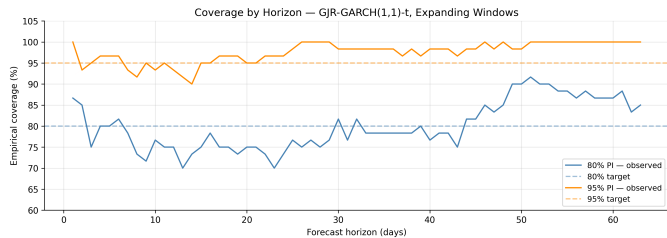


Fig. 10. Empirical coverage of GJR-GARCH(1,1)-t prediction intervals across 60 expanding-window origins, by horizon step. Solid lines: observed coverage; dashed: nominal 80% and 95% targets. Both intervals meet or slightly exceed nominal across the full 63-day horizon.

TABLE III
GJR-GARCH(1,1)-T SPECIFICATION CHOICES

Decision	Justified by
Model log-returns, not prices	ADF, KPSS, PP (Sec. VII-A)
Constant mean, not AR	HAC t -stat $ t = 0.74$, $p = 0.46$
Student- t , not Normal	Excess kurtosis 6.65; Jarque-Bera
GARCH family over IID	ARCH LM, $p \approx 5 \times 10^{-106}$
GJR over plain GARCH	AIC, BIC, squared-residual LB
Expanding-window fits	Stabilizes ν ; calibrated coverage
80% / 95% PIs	Empirical coverage at or above nominal

Expanding rather than rolling windows stabilizes the Student- t degrees-of-freedom estimate, which is notoriously sample-sensitive; across the 60 origins ν has median 4.79, range [4.18, 5.64], and standard deviation 0.43. At each origin we produce 63-step forecasts of the conditional variance, integrate to cumulative variance, and back out 80% and 95% prediction intervals on prices using the Student- t quantiles. We then count the fraction of realized prices that fall inside each band at each step.

Fig. 10 shows the result. Empirical 80% and 95% coverage track their nominal targets across the full horizon, both sitting at or slightly above the dashed reference lines. The intervals are mildly conservative rather than under-covering, which is the safe direction for a deployment forecast.

Fig. 11 provides a qualitative complement: six origins drawn from the 60, each showing the realized path against the point forecast and the 80% and 95% intervals. The realized path stays inside the 95% band at all six origins and inside the 80% band most of the time, including across the high-volatility 2011 episode (Origin 26) and the 2024–2025 rally (Origin 59).

H. Summary

Table III collects the specification choices and the test that justifies each.

I. Scope of the specification search

GJR-GARCH(1,1)-t is presented here as an *informed* choice within the broader GARCH family rather than the optimum of an exhaustive specification search. We compared the chosen model against plain GARCH(1,1)-t and verified it on

the standard diagnostic ladder (drift, distribution, ARCH effects, asymmetry, residual autocorrelation, out-of-sample coverage), but did not evaluate alternative volatility families such as EGARCH, FIGARCH, component GARCH, stochastic-volatility, or HAR-RV models. A more thorough specification search may yield marginal improvements in coverage or tightness; the chosen specification is the one that satisfies the in-sample diagnostics and the out-of-sample 80% / 95% coverage targets the procurement use-case requires, with the lowest complexity that does so.

IX. DEPLOYMENT

The GJR-GARCH(1,1)-t model is operationalized as a daily forecast running on a scheduled GitHub Actions cron. Each invocation loads the committed historical series, appends any newer ICE *Coffee C* closes from Yahoo Finance (ticker KC=F), refits the model on log-returns, and writes a 63-day forecast with 50%, 80%, and 95% intervals to a CSV plus a PNG plot. The latest run is held at a fixed path and each daily run is archived in a year/month-organized directory for traceability. The pipeline is reproducible from a fresh clone; full source, the model package, and the four analysis notebooks are publicly available [36].

Fig. 12 shows the production forecast at the time of writing: the recent observed price path, the GJR-GARCH(1,1)-t point forecast, and the 50%, 80%, and 95% prediction intervals over the 63-day horizon. The image is regenerated by every scheduled run and serves as the visible artifact of the deployed system.

X. CONCLUSION

We benchmarked ten forecasting models spanning five methodological categories on 32 years of ICE *Coffee C* nearby-close daily prices. The 60-origin rolling-window back-test, the Diebold-Mariano test against a random-walk baseline, and the Hansen-Lunde-Nason Model Confidence Set all converge on the same finding: nine of the ten models are jointly indistinguishable in terms of point-forecast accuracy at $\alpha = 0.10$, including a foundation model with a 32-year pretrained backbone. Only Prophet falls outside the confidence set.

The a priori forecastability battery explains why. ADF, KPSS, and Phillips-Perron place the series on a unit root; the heteroskedasticity-robust Lo-MacKinlay variance ratio fails to reject the random-walk null at every tested horizon; the Hurst exponent is just above 0.5 but Lo’s modified V_n does not reject non-long-memory; spectral predictability on log-returns is near the white-noise floor; permutation entropy is near the ceiling on both representations; and the raw Lyapunov exponent is statistically indistinguishable from that of shuffled returns. The empirical signature is consistent with weak-form market efficiency.

Because the benchmark’s point-forecast accuracies are indistinguishable, the deployment model is chosen on the basis

GJR-GARCH(1,1)-t sample forecasts across six historical origins

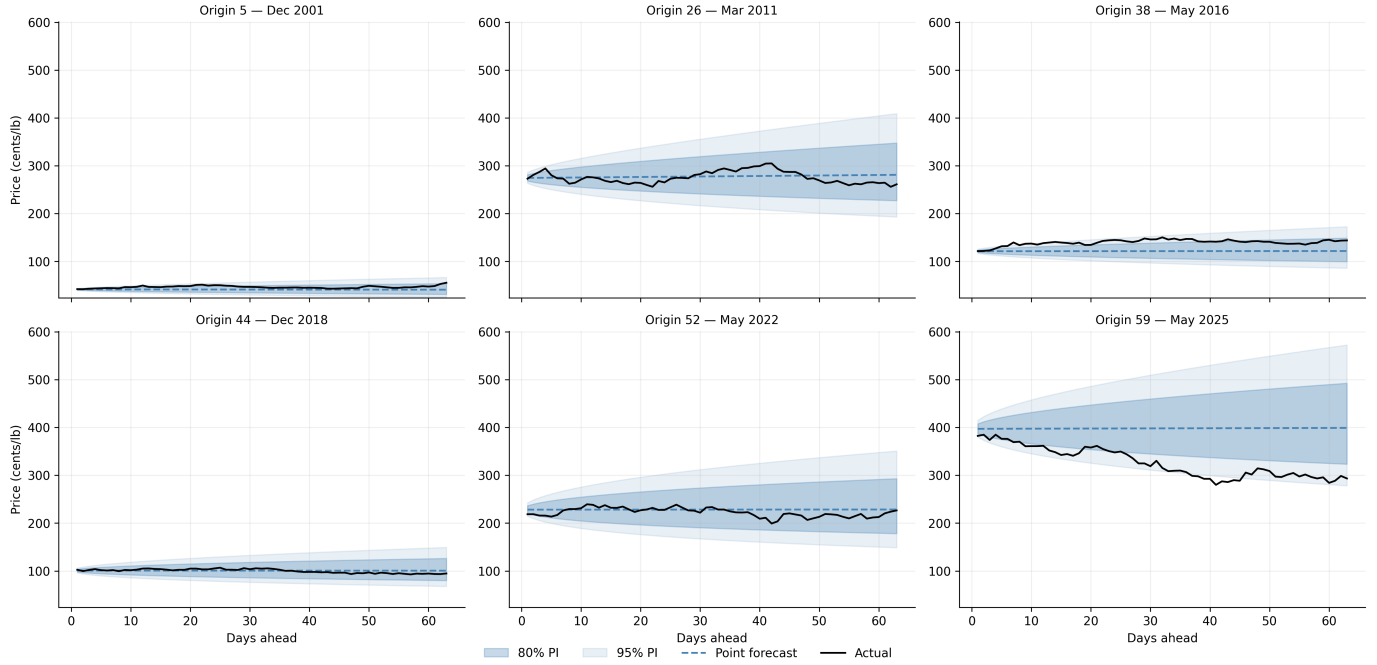


Fig. 11. Six sample GJR-GARCH(1,1)-t forecasts drawn from the 60-origin backtest. Each panel shows the realized 63-day path (black) against the point forecast (dashed) and the 80% and 95% prediction intervals (shaded), produced by an expanding-window refit at the indicated origin date.

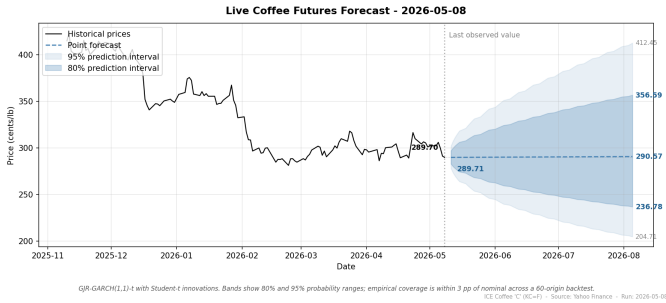


Fig. 12. Production output of the deployed pipeline at the time of writing: ICE Coffee C nearby-close history (left of the vertical separator), the GJR-GARCH(1,1)-t point forecast, and the 50%, 80%, and 95% prediction intervals over the next 63 trading days. The image is overwritten on every scheduled run of the daily pipeline.

of prediction intervals. A GJR-GARCH(1,1)-t model with constant mean and Student- t innovations is justified specification-by-specification through HAC-robust drift tests, the ARCH LM test, AIC/BIC comparisons against plain GARCH, and Ljung-Box on squared standardized residuals. An expanding-window backtest at 60 historical origins confirms calibrated empirical coverage at the nominal 80% and 95% levels across the full 63-day horizon. The deployed system produces a daily forward distribution that is appropriate for the procurement use-case that motivated the project.

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